HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **PLC and HMI applications** for automated process equipment, troubleshooting and resolving issues across instrumentation and controls.
- Managed projects focused on the design, commissioning, and performance enhancement of **automated equipment** within a regulated manufacturing environment.
- Implemented software and hardware changes to automated systems, meticulously documenting and assessing the impact on manufacturing and facilities operations.
- Contributed to the design and approval of hardware, software, and controls for new automated equipment, ensuring alignment with project specifications.
- Supported the Validation Department through initial design reviews, comprehensive software testing, and protocol execution and analysis.
- Advanced modification and change control procedures and protocols for automated systems, ensuring compliance and traceability.
- Containerized and deployed **Java Spring Boot** microservices to an AWS Elastic Kubernetes Service (EKS) cluster, managing configuration and resource allocation.

SKILLS

Automation & Controls – PLC Programming, HMI Development, Instrumentation, Process Equipment, Automated Systems

Software Development – Java, Spring Boot, REST APIs, Docker, Kubernetes, EKS, Python

Project Management & Support – Project Design, Commissioning, Performance Improvement, Technical Support, Change Control, Validation Support

Manufacturing & Operations – Manufacturing Operations, Facilities Support, Regulated Environment, Software Testing, Protocol Execution

Data & Analytics – Power BI, Tableau, DAX, Power Query, SQL, R, pandas, scikit-learn

PROJECTS

Automated Pharmaceutical Manufacturing Line Integration

- Designed and implemented **PLC and HMI logic** for a new automated packaging line, integrating with existing MES systems to ensure seamless data flow.
- Developed Python scripts to automate the generation of **commissioning reports** and performance validation summaries, reducing manual effort by 40%.
- Collaborated with cross-functional teams, including process engineers and quality assurance, to define requirements and ensure successful system integration.

Java Backend for Manufacturing Execution System (MES)

- Developed **Java Spring Boot** microservices to manage real-time data ingestion from shop floor equipment, serving critical operational metrics.
- Implemented a **PostgreSQL** persistence layer for storing production batch data, designing schemas to support historical trend analysis and reporting.
- Containerized the services using **Docker** and deployed to an AWS environment, establishing CI/CD pipelines with Jenkins for automated builds and deployments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)